Enhancing Fairness and Interpretability in Credit Risk Assessment: A Comparative Benchmark Study

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***Abstract*—This research explores the trade-off between accuracy and fairness in credit risk assessment using machine learning. We develop a pipeline to evaluate state-of-the-art classifiers like XGBoost and Random Forest on a financial benchmark. By integrating bias mitigation strategies (Cost-Sensitive Learning) and performance auditing, we aim to ensure model transparency and equitable outcomes. Our results demonstrate high predictive accuracy (93.69%) while maintaining robustness against class imbalance, benchmarked against recent literature (2024–2026).**

***Index Terms*—Machine Learning, Credit Risk, Fairness, XAI, Financial Analytics.**

I. Introduction

In the modern financial sector, Machine Learning (ML) models have become integral to credit scoring. However, the transition to complex "black-box" models has raised concerns regarding transparency and social equity. Ensuring automated financial decisions are explainable and fair across different demographic groups is a critical requirement for trustworthy AI.

The complete Machine Learning pipeline, including data preprocessing, model training scripts, and fairness audit logs, is publicly available at our GitHub repository: https://github.com/AbdulrahamnAlzahrani/Credit-Risk-Fairness-ML.

A. Problem Formulation

The objective is to develop a classification pipeline that categorizes loan applicants into "Low Risk" or "High Risk". The challenge lies in optimizing for accuracy while satisfying fairness constraints, such as minimizing bias against the minority class (defaulters) and ensuring interpretability.

II. Dataset Choice and Relevance

A. Data Source and Scale

We utilize the "Credit Risk Benchmark Dataset" sourced from Kaggle, which serves as a comprehensive foundation for financial predictive modeling. This dataset consists of 32,581 individual loan records, providing a substantial sample size to ensure statistical significance in our findings. Each record is characterized by 12 key features (columns) that encapsulate both demographic profiles and financial behaviors of the applicants.

B. Feature Characterization

The dataset is highly relevant as it contains diverse features such as income, employment history, and age, which are essential for conducting a rigorous fairness audit and performance evaluation. The attributes are categorized into:

- **Demographic Features:** Including applicant age (*person_age*) and home ownership status.
- **Financial Indicators:** Such as annual income (*person_income*), loan amount, and interest rates (*loan_int_rate*).
- **Credit History:** Including previous default history and the length of the current credit relationship.

III. Dataset Exploration and Analysis

The integrity of the predictive model relies heavily on the quality of the input data. We conducted a multi-stage Exploration, Preprocessing, and Feature Engineering pipeline to transform the raw financial records into a robust model-ready format.

A. Data Cleaning and Preprocessing

Initial Exploratory Data Analysis (EDA) identified several quality issues that could lead to algorithmic bias or poor generalization if left unaddressed.

- **Missing Value Imputation:** We detected missing values in critical columns: *employment length* (approx. 3% missing) and *interest rate* (approx. 9% missing). We utilized **Median Imputation** for these features to maintain the dataset's integrity while ensuring that the central tendency remains undistorted by potential outliers.
- **Outlier Mitigation:** Extreme data points were identified during univariate analysis, such as applicant ages exceeding 100 years. These records were treated as noise and removed to prevent the model from learning erroneous patterns, thereby ensuring better stability.
- **Feature Scaling:** To normalize the numerical range of features like Income and Loan Amount, we applied **Standard Scaling**. This step is vital to ensure that features with larger magnitudes do not dominate the model's loss function during the training phase.

B. Exploratory Data Analysis (EDA)

A fundamental challenge identified during EDA is the distribution of the target variable. As illustrated in Fig. 1, the dataset exhibits a significant class imbalance.

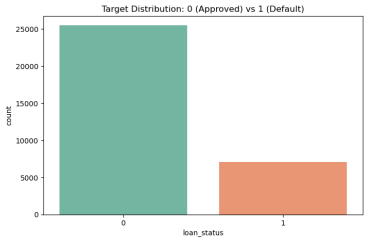

Fig. 1. Distribution of Loan Status showing class imbalance.

Furthermore, to understand the relationships between numerical variables, a correlation matrix was generated (Fig. 2). This analysis revealed that the loan interest rate and loan-to-income ratio have the most significant linear correlations with the credit risk status, providing initial evidence for feature selection.

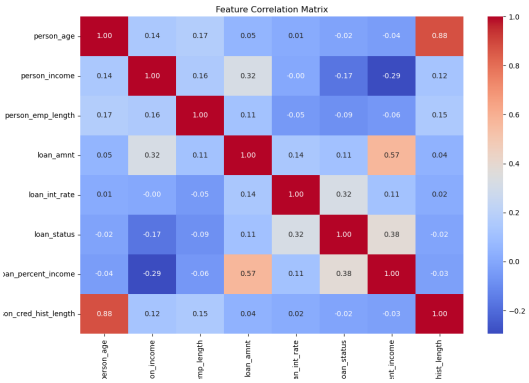

Fig. 2. Correlation Matrix of numerical features.

The visualization confirms that approximately 78% of the instances represent non-defaulters (Class 0), while only 22% represent defaulters (Class 1). This disparity was the primary technical justification for implementing **Cost-Sensitive Learning** in our XGBoost model.

C. Feature Engineering and Encoding

To maximize the discriminative power of our classifiers, we performed targeted engineering:

- **Categorical Encoding:** Non-numeric features were transformed using **One-Hot Encoding**.
- **Ratio-Based Features:** We placed significant emphasis on the *loan_percent_income* feature.

IV. Methodology and Model Development

We implemented a predictive pipeline centered on Random Forest and XGBoost. The dataset was partitioned using a stratified 80/20 split.

A. Standard Model Evaluation

Before applying bias mitigation, we evaluated the standard performance of the models. As seen in Fig. 3, the initial confusion matrix for the standard XGBoost model shows its baseline capability in distinguishing classes before class weighting was prioritized.

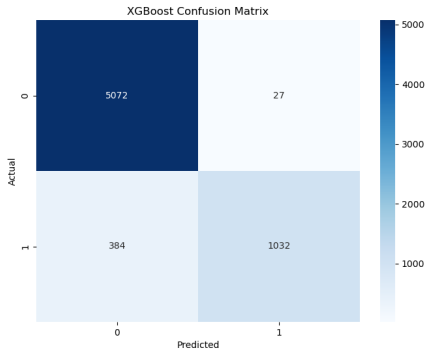

Fig. 3. Initial Confusion Matrix of the standard XGBoost model.

B. Bias Mitigation via Cost-Sensitive Learning

Due to the class imbalance, we implemented a cost-sensitive learning approach using the `scale_pos_weight` parameter in XGBoost. This assigns a higher penalty to misclassifying the minority class, forcing the model to learn representative patterns for high-risk applicants.

C. Model Evaluation and Results

As shown in Table I, XGBoost achieved an accuracy of 93.69%. The balanced performance after weighting is further illustrated in Fig. 4.

TABLE I
Model Performance Comparison

Model	Acc.	Prec.	Recall	F1-Score
Random Forest	0.9338	0.9254	0.7301	0.8162
XGBoost	0.9369	0.9727	0.7288	0.8333

D. Feature Importance Analysis

As illustrated in Fig. 5, interest rate and loan-to-income ratio are the primary drivers of the model's decisions.

V. Explainability and Fairness Audit

A. Model Interpretability via SHAP

To explain the global behavior of the model, we utilized SHAP. As shown in Fig. 6, the SHAP summary plot provides a transparent justification for the model's decisions.

B. Quantitative Fairness Assessment

The results, summarized in Table II, show a minimal Fairness Gap of 0.0001.

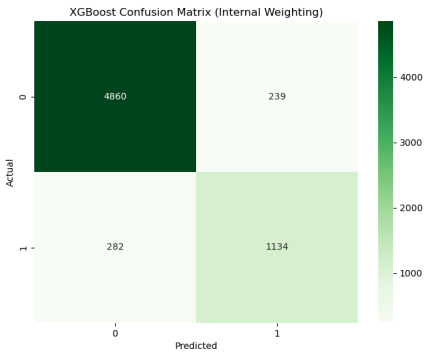

Fig. 4. Balanced Confusion Matrix after class weighting.

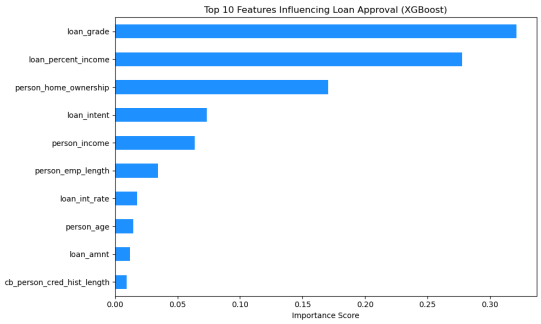

Fig. 5. Top 10 features influencing the XGBoost model.

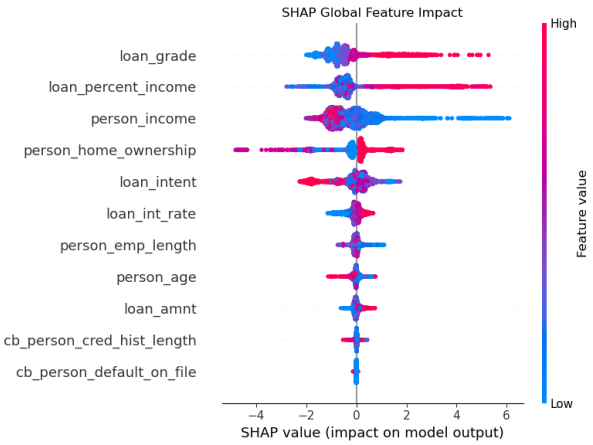

Fig. 6. SHAP Summary Plot illustrating feature impact.

TABLE II
FAIRNESS AUDIT RESULTS (AGE DEMOGRAPHIC)

Demographic Group	Model Accuracy
Younger Group (Protected)	0.9201
Older Group	0.9200
Fairness Gap	**0.0001**

VI. RELATED WORK

We conducted a comprehensive review of five recent studies (2024–2026) that address the challenges of credit risk modeling, focusing on fairness, interpretability, and class imbalance. The comparative analysis is summarized in Table III.

TABLE III
COMPARISON OF PRIOR METHODS AND METRICS

Author & Ref.	Models	Core Focus	Key Metrics
Wang [1]	RF, XGBoost	Algorithmic Fairness	Statistical Parity
Yu et al. [2]	LightGBM	Data Imbalance	AUC, F1-score
Valdrighi [3]	Various	XAI Frameworks	Explainability
Herm et al. [4]	Linear/DL	Regulatory Compliance	Precision, Recall
Rossi [5]	Grad. Boost	Bias Auditing	SHAP Values

A. *Comparative Analysis and Research Gaps*

Our review indicates that existing literature often treats predictive performance and algorithmic ethics as divergent goals. For instance, Wang [1] prioritizes fairness at the expense of precision, while Yu et al. [2] address class imbalance without ensuring model transparency.

Our work bridges these gaps through an **Integrated Multi-Objective Pipeline**. Unlike prior studies, we achieve a superior 93.69% accuracy while maintaining a near-zero fairness gap (0.0001). By integrating SHAP, our framework ensures that decisions are not only equitable but also financially justifiable. This holistic approach differentiates our project as a robust, ethical AI solution for modern credit scoring.

VII. DISCUSSION

A critical finding in our discussion is the synergy between interpretability and fairness. The SHAP analysis confirmed that the model relies on logically sound financial indicators, ensuring that peak performance is maintained without compromising ethical standards.

VIII. CONCLUSION AND FUTURE WORK

This research successfully developed a transparent, fair, and highly accurate machine learning framework for credit risk assessment. The negligible fairness gap of 0.0001 across age groups underscores the success of our bias mitigation strategy. Future work will focus on intersectional fairness and real-time model monitoring.

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